

Supplementary Information: Rapid Epitope-Targeted Nanobody Design via Differentiable Distogram Optimization with ESMFold2

1 Supplementary Tables

Table S1: Parameter sweep results Systematic parameter sweep on RBD/VHH72 (6WAQ). All runs use ESMFold2-Fast (721M) on MPS, single seed. “Bounce” indicates cdr→epitope oscillation magnitude.

The optimal parameter set ($wt_logit = 2.0$, $w_{prior} = 0.05$, 60 steps) was used for all subsequent design experiments.

Table S2: Complete design results (15 designs, 3 seeds $\times$ 5 targets)

Table S3: Best CDR sequences per target

2 Supplementary Figures

Figure S1: Fast vs Full model comparison ESMFold2-Fast (721M) and full ESMFold2 (1.3B) show systematic discrepancies in cdr→epitope distance estimates. Fast model values are uncalibrated and should be used only for relative trends during design, with full model evaluation for final ranking.

[htbp]

Table 1: Parameter sweep on RBD/VHH72. wt_logit is the key bottleneck.

wt_logit	w_{prior}	Steps	Final cdr→epi (Å)	Best (Å)	Bounce
5.0	0.30	30	17.3	9.6	Severe
2.0	0.30	30	15.3	8.9	Moderate
1.5	0.30	30	15.5	11.6	Moderate
1.0	0.30	30	11.3	10.4	Mild
1.0	0.30	60	11.0	10.6	Mild
0.5	0.30	60	17.3	14.7	Severe
1.0	0.10	60	13.9	12.5	Mild
1.5	0.05	60	13.0	11.5	Moderate
2.0	0.05	60	9.3	9.3	None

[htbp]

Table 2: Complete results for all 15 design seeds. Evaluation: Full ESMFold2, 3 loops, 14 sampling, ds=1.

Target	Seed	ipTM	pTM	cdr→epi (Å)	Mutations
3*Ty1/RBD (6ZXN)	0	0.471	0.735	10.7	10/22
	1	0.702	0.823	9.2	11/22
	2	0.546	0.765	11.3	15/22
3*KN035/PD-L1 (5JDS)	0	0.447	0.680	11.9	12/32
	1	0.459	0.690	11.5	16/32
	2	0.449	0.679	12.8	11/32
3*VHH3/TNF α (5M2M)	0	0.671	0.795	11.6	17/33
	1	0.654	0.786	17.6	15/33
	2	0.653	0.784	15.3	19/33
3*VHH72/RBD (6WAQ)	0	0.742	0.857	10.0	11/29
	1	0.698	0.832	9.6	11/29
	2	0.676	0.819	13.9	15/29
3*anti-TNF (5M2J)	0	0.110	0.531	18.3	12/19
	1	0.231	0.568	15.4	10/19
	2	0.083	0.509	18.3	11/19

[htbp]

Table 3: Wild-type and best design CDR sequences for each target.

Target	Type	CDR Sequence
2*Ty1/RBD	WT	GFTFSSVSPNSGNLNLSSSSV
	Des	GFTLANHNPESGMGLNLAVSPD
2*KN035/PD-L1	WT	GKMSSRRLTTSGSDSFEDPTCTLVTSSGAFQY
	Des	GMMSSRRLTTSGSDQFGLPTCDHVNSSGAFQY
2*VHH3/TNF α	WT	GRTFSDHSGYTYWSSGNRDGIPTSRSVESYNY
	Des	GRKFSKTPKYTNYISSVQITGIPKKSSYELYNT
2*VHH72/RBD	WT	GRTFSEYSWSGGSAGLGTVVSEWDYDYDY
	Des	SPKFKDVLMSAGRAGLGTVVSMWDYDYDY
2*anti-TNF	WT	GFTFSNYNTNGLISPSGFN
	Des	GKKFAAYNTPGFQDPSGVD

	Target	Fast cdr→epi (WT)	Full cdr→epi (WT)	Gap
[htbp]	RBD/VHH72	16.8 Å	9.6 Å	7.2 Å
	Ty1/RBD	17.5 Å	16.3 Å	1.2 Å
	PD-L1/KN035	N/A	14.1 Å	—

Figure S2: Per-target detailed views [htbp]

Figure S3: Method overview schematic The three-stage pipeline: (1) Full ESMFold2 predicts CA-coordinate structure prior; (2) ESMFold2-Fast serves as differentiable oracle for CDR logit optimization via gradient descent through the distogram; (3) Full ESMFold2 evaluates final candidates.

3 Supplementary Methods

Extended loss function details The complete loss function is:

$$\mathcal{L}_{\text{total}} = \sum_k w_k \mathcal{L}_k + w_{\text{aa}} \mathcal{L}_{\text{aa_freq}}, \quad (1)$$

where $k \in \{\text{intra_contact}, \text{inter_contact}, \text{glob}, \text{epitope}, \text{structure_prior}\}$.

Intra-contact loss: Binder internal contact prediction entropy, encouraging well-defined intra-domain contacts. **Inter-contact loss:** Binder-target interface contact prediction entropy, encouraging compact interfaces. **Global loss:** Global structure quality regularization. **Structure prior loss:** Masked cross-entropy between predicted distogram and pre-computed CA prior.

The prior is computed from full ESMFold2 CA coordinates averaged over 4 diffusion samples. CA distances are discretized into 64 bins spanning 2 Å to 22 Å, with a bin tolerance of 2.5 Å.

ESMFold2 pose basin analysis Our earlier investigation revealed that ESMFold2’s predicted complex pose is a deterministic function of the framework sequence, not a sample from a distribution. This was confirmed by: (1) multiple independent folds of the same sequence producing poses within 1.84 Å binder RMSD of each other; (2) injecting a 30°-rotated binder pose at the diffusion start having no effect on the final output; (3) comparing framework-variant vs. CDR-variant predictions, where framework changes shift the pose by 0.29 Å but CDR changes produce 2.54 Å variation. This finding underlies our strategy: we keep the framework fixed and optimize only CDRs.

Figure 1 Fast vs. Full ESMFold2 cdr→epitope distance comparison.

Figure 2 Per-target design trajectory (left) and random baseline distribution (right) for Ty1/RBD and KN035/PD-L1.